Part III

Finite Element Methods

Chapter 10

Elliptic Problems

10.1 Preliminary Remarks

Mathematical models of a variety of physical processes lead to second order equations of the type

$$-\sum_{i,j=1}^{d}(a_{ij}(x)u_{x_i})_{x_j}+\sum_{i=1}^{d}(b_i(x)u)_{x_i}+d(x)u=f(x), \qquad x=(x_1,x_2,\ldots,x_d)^T\in\Omega\subset\mathbb{R}^d \quad (10.1)$$

where the coefficients a_{ij}, b_i, d, and f are functions of x, or simply constants. In the latter case, eq. (10.1) is identical to (B.2) in Appendix B.1, which we have used to classify PDE.[1] Recall that we that we say a second order PDE is *elliptic*, if the eigenvalues of the matrix defined by $(a_{ij})_{i,j=1,\ldots,d}$ all have the same sign. If the PDE is elliptic, we may assume, without loss of generality, that all eigenvalues are positive. (If all eigenvalues are negative, we may multiply the equation by -1 to obtain a coefficient matrix with positive eigenvalues.)

In the general case, where the coefficient matrix $A=(a_{ij})_{i,j=1,\ldots,d}$ may depend on x, the eigenvalues must be positive for *any* x in the domain of interest. The following, slightly stronger definition of *uniform* ellipticity will prove useful:

Definition 10.1 *We call the differential operator L with*

$$Lu:=-\sum_{i,j=1}^{d}(a_{ij}(x)u_{x_i})_{x_j}+\sum_{i=1}^{d}(b_i(x)u)_{x_i}+d(x)u \quad (10.2)$$

uniformly *elliptic, if there is a positive constant α' such that*

$$\xi^T A(x)\xi=\sum_{i,j=1}^{d}\xi_i a_{ij}(x)\xi_j\geq\alpha'|\xi|^2 \quad (10.3)$$

for all $\xi\in\mathbb{R}^d$ and $x\in\Omega$. Here $|\xi|^2:=\xi_1^2+\cdots+\xi_d^2$ is the usual Euclidean vector norm.

Obviously this implies that the matrix A is positive definite, which in turn implies that all the eigenvalues are strictly positive. The significance of Definition 10.1 is that the eigenvalues of

[1] Note that even if the coefficients are not constants, but smooth functions, we can differentiate using the product rule, and re-write (10.1) in the form (B.2), where we replace b_i with $\widetilde{b}_i=b_i-\sum_j(a_{ij})_{x_j}$, and d with $\widetilde{d}=d+\sum_i(b_i)_{x_i}$. The form (10.1) is sometimes called the *divergence form*.

A are uniformly bounded away from zero, i.e. by a distance of α'. We shall see in later sections that such a uniform bound is enormously useful in proving stability of numerical approximations.

Since the ellipticity of (10.1) depends only on the matrix A, we assume for the moment $b_i = d = 0$, and consider the differential equation

$$-\sum_{i,j=1}^{d} (a_{ij}(x)u_{x_j})_{x_i} = f(x). \tag{10.4}$$

We can write (10.4) more succinctly as

$$-\nabla \cdot (A\nabla u) = f. \tag{10.5}$$

(Here we have suppressed the x-dependence of the coefficients in the notation, which we shall do often.) Of course we need to augment problem (10.4) with proper boundary conditions and, perhaps, initial conditions. It is not clear yet, whether one of the d coordinates represents time. For simplicity, we consider the case $A = I$, where I is the identity matrix. The problem then reduces to the *Poisson equation*.

$$-\nabla^2 u = f,$$

where the Laplacian operator ∇^2 with

$$\nabla^2 u := \nabla \cdot (\nabla u) = \sum_{i=1}^{d} u_{x_i x_i}$$

has been defined. For the moment, we set $f = 0$ (the equation is then called the *Laplace Equation*), and investigate whether the elliptic problem is well posed as a Cauchy problem, similar to that investigated in appendix B.3, eq. (B.9)-(B.11), in the context of the wave equation.

$$\begin{aligned} u_{tt} + u_{xx} &= 0, & x \in \mathbb{R},\ t \in (0,T], \\ u(x,0) &= u_0(x), & x \in \mathbb{R}, \\ u_t(x,0) &= 0, & x \in \mathbb{R}. \end{aligned}$$

The Laplace equation bears a resemblance to the wave equation, in that there is only a sign difference between one of the terms that are being differentiated. This difference turns out to be rather profound, however. It is responsible for the wave equation being hyperbolic, and the Laplace equation being elliptic. We show that the time-dependent Cauchy problem for the Laplace equation is *not* in general well posed. We use an approach similar to that used in Example 2.1 for the heat equation, and consider initial data

$$u_0^{(k)}(x) = \frac{1}{k}\sin(kx), \qquad k \in \mathbb{N}.$$

Straightforward differentiation reveals that

$$u^{(k)}(x,t) = \frac{1}{k}\cosh(kt)\sin(kx) \tag{10.6}$$

is a solution[2]. For a well-posed problem the solution is bounded by the data, and consequently $u^{(k)}$ is bounded by the perturbation in the initial data. In particular, this means that $u^{(k)}$ must become arbitrarily small, as k grows large. To show that this is *not* the case, consider the solution at time $t = T$.

$$\begin{aligned} \max_x |u^{(k)}(x,T)| &= \max_x \left| \frac{1}{k}\cosh(kT)\sin(kx) \right| \\ &= \frac{1}{2k}(e^{kT} + e^{-kT}) \end{aligned}$$

Obviously then, the changes in the solution become arbitrarily large, as k grows, while the amplitude of the perturbation becomes smaller. Hence the problem is not well posed.

[2]Note: $\cosh(x) = \frac{1}{2}(e^x + e^{-x})$.

Remark 10.1 The previous statement on well posedeness can be extended to initial-boundary value problems. The solution (10.6) given above actually satisfies the initial-boundary value problem

$$\begin{aligned} u_{tt} + u_{xx} =& 0 && x \in (0,\pi),\ t \in (0,T], \\ u(x,0) =& \frac{1}{k}\sin(kx) && x \in [0,\pi], \quad k \in \mathbb{N}, \\ u_t(x,0) =& 0, && x \in [0,\pi], \\ u(0,t) = u(\pi,t) =& 0, && t \in (0,T]. \end{aligned}$$

which is also not well posed.

From now on we shall only consider steady elliptic problems. Such problems are characterized by absence of a "time" variable. We discuss boundary conditions for such problems as we progress.

10.2 Finite Difference Method for the Poisson Problem

Consider first the Poisson equation on a simple square domain $\Omega = (0,1) \times (0,1)$:

$$\begin{aligned} -\nabla^2 u = f, \qquad & (x,y) \in \Omega \\ u(x,y) = 0 \qquad & (x,y) \in \partial\Omega \end{aligned}$$

where f is some given function. Using our expertise in difference approximations it is easy to devise a difference scheme for this equation. Recall that for any smooth function

$$\begin{aligned} u_{xx} &= \frac{u(x+h,y) - 2u(x,y) + u(x-h,y)}{h^2} + \mathcal{O}(h^2) \\ u_{yy} &= \frac{u(x,y+h) - 2u(x,y) + u(x,y-h)}{h^2} + \mathcal{O}(h^2) \end{aligned}$$

To construct a numerical scheme based on this difference approximation, we define a grid

$$\mathcal{G}_h := \{(ih, jh) : i,j = 0,\ldots,N, \quad hN = 1\}, \tag{10.7}$$

and the scheme

$$\begin{aligned} -\frac{1}{h^2}\left(u_{i-1,j} - 2u_{i,j} + u_{i+1,j}\right) - \frac{1}{h^2}\left(u_{i,j-1} - 2u_{i,j} + u_{i,j+1}\right) &= f_{i,j}, \qquad i,j = 1,\ldots N-1 \\ u_{0,j} = u_{N,j} &= 0, \qquad \text{all } j \\ u_{i,0} = u_{i,N} &= 0, \qquad \text{all } i \end{aligned}$$

This leads to a large linear system of equations,

$$A\mathbf{u} = \mathbf{f}, \tag{10.8}$$

where we may write the resulting matrix as a block-matrix

$$A = \frac{1}{h^2}\begin{pmatrix} D & -I & & & & \\ -I & D & -I & & & \\ & -I & D & -I & & \\ & & & \ddots & & \\ & & & -I & D & -I \\ & & & & -I & D \end{pmatrix}$$

where I is the $(N-1)\times(N-1)$ identity matrix, and $D \in \mathbb{R}^{(N-1)\times(N-1)}$ is given by

$$D = \begin{pmatrix} 4 & -1 & & & & \\ -1 & 4 & -1 & & & \\ & -1 & 4 & -1 & & \\ & & & \ddots & & \\ & & & -1 & 4 & -1 \\ & & & & -1 & 4 \end{pmatrix},$$

while

$$\mathbf{u} = \begin{pmatrix} u_{1,1} \\ u_{2,1} \\ u_{3,1} \\ \vdots \\ u_{N-1,1} \\ u_{1,2} \\ u_{2,2} \\ u_{3,2} \\ \vdots \\ u_{N-3,N-1} \\ u_{N-2,N-1} \\ u_{N-1,N-1} \end{pmatrix} \qquad \mathbf{f} = \begin{pmatrix} f_{1,1} \\ f_{2,1} \\ f_{3,1} \\ \vdots \\ f_{N-1,1} \\ f_{1,2} \\ f_{2,2} \\ f_{3,2} \\ \vdots \\ f_{N-3,N-1} \\ f_{N-2,N-1} \\ f_{N-1,N-1} \end{pmatrix}$$

We note that the matrix A is symmetric positive definite. Also note that the structure of the matrix depends on the ordering that we have chosen for our unknowns $\mathbf{u}$.

Devising suitable difference schemes does not seem like a big challenge. The only complication could result from the fact that the systems we are faced with can become quite large. Rather than using a direct method to solve them, we may consider iterative schemes.

It is worth noting a few comments on finite-difference schemes for the elliptic problem. It is obvious how they can become quite cumbersome for complicated computational domains, and furthermore, they require rather high regularity to obtain $\mathcal{O}(h^2)$ error estimates. (From Taylor series analysis it is evident that the solution must be four times continuously differentiable in order for the Taylor series analysis to make sense.) We shall see in the next few chapters that the finite element method is much more flexible, and allows error estimates using weaker assumptions.

10.3 Further Reading

Evans [4], Chapter 2, gives a proof of the uniqueness of the (strong) solution to the Poisson problem using a maximum principle. We skip this here, since we re-visit the question of well-posedness for the more general weak solution. The example showing that the Laplace equation is not well-posed as a time-dependent problem has been taken from Braes [3], Chapter 1.